Сырокв Р
ИВТ 2-1

2)

$$U_k = (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

$$M = \frac{U_k}{U_{k-1}} = \frac{(-1)^k \frac{x^{2k+1}}{(2k+1)!}}{(-1)^{k-1} \frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} =$$

$$= \frac{(-1)\cancel{(-1)^{k-1}} \frac{x^{2k+1}}{(2k+1)!}}{\cancel{(-1)^{k-1}} \frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} = (-1) \cdot \frac{x^{2k+1}}{(2k+1)!} \cdot$$

$$\frac{(2(k-1)+1)!}{x^{2(k-1)+1}} = (-1) \frac{x^{2k} \cdot \cancel{x} (2(k-1)+1)!}{(2k+1)! \, \cancel{x^{2(k-1)}} \cdot \cancel{x} x^{2k}} \cdot \frac{1}{x^2}$$

$$= (-1) \frac{x^2 (2k-1)!}{(2k+1)!} = \cancel{(-1) \frac{x^2 (2k-1)!}{2k(2k+1)(2k-1)!}}$$

$$(-1) \cdot \frac{x^2}{4k^2+2k}$$

$$M = -\frac{x^2}{4k^2+2k}$$

3) $U_k = (-1)^k \cdot \frac{x^{2k}}{(2k)!}$

$$M = \frac{U_k}{U_{k-1}} = \frac{(-1)^k \cdot \frac{x^{2k}}{(2k)!}}{(-1)^{k-1} \cdot \frac{x^{2(k-1)}}{(2(k-1))!}} =$$

$$= \frac{-1 \cdot \frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)!}} = \frac{-\frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)}} =$$

$$= -\frac{\frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)!}} = -\frac{x^2 \cdot (2k-2)!}{(2k)!} =$$

$$= -\frac{x^2 \cdot (2k-2)!}{2k \cdot (2k-1) \cdot (2k-2)!} = \frac{x^2}{2k \cdot (2k-1)} =$$

$$= -\frac{x^2}{4k^2-2k}$$

$$\underline{M = -\frac{x^2}{4k^2-2k}}$$